Task 2 -- Baseline Benchmarking: Base SLM vs. Reference LLM

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| Field | Value | |---|---| | **Notebook** | 02 of 05 -- Task 2: Baseline Benchmarking (1.5 marks) | | **Systems compared** | (A) base SLM -- no fine-tuning | (B) reference LLM -- no fine-tuning | | **Models** | (A) `Qwen/Qwen2.5-1.5B-Instruct` | (B) `Qwen/Qwen2.5-7B-Instruct` (4-bit inference) |

MUST RUN IN THE BITS VIRTUAL LAB. This notebook downloads and runs two language models and therefore needs GPU memory (recommend >=8 GB free VRAM for the 7B reference model in 4-bit, plus ~3-4 GB for the 1.5B SLM in bf16). It was authored and syntax-validated outside the Virtual Lab (no GPU available there) but **all outputs must be produced by actually executing it in the Virtual Lab**, per PS2's submission rules. Take full-screen screenshots while it runs, per the assignment's Virtual Lab proof requirement.

Objective: establish the starting gap between the small and large models on a frozen 8-prompt benchmark, before any fine-tuning -- this gap is exactly what Task 3/4's fine-tuning must close, and this same benchmark is reused verbatim in Notebooks 04 and 05.

Problem Understanding & Theory

Why a *frozen* benchmark? If the evaluation prompts changed between Tasks 2, 4 and 5, any apparent improvement could just be "easier prompts this time" rather than genuine model improvement. Freezing the exact 8 prompts (`data/benchmark/frozen_benchmark_prompts.json`, built in Notebook 01's sibling step) makes every later comparison a controlled, apples-to-apples measurement.

Why these 8 prompt types? PS2 requires: in-domain factual questions, one formatting-sensitive instruction, one reasoning-style question, and one out-of-scope request. The frozen set (see table below) covers 5 in-domain factual prompts (broad intent coverage: card activation, loan documents, ATM card retention, credit-card block-vs-cancel, ATM dispute), 1 strict-format instruction (numbered list, exactly 4 steps -- this is where base SLMs typically fail even when the *content* is right), 1 multi-step reasoning/arithmetic question (balance + transfer + bill payment ordering), and 1 clearly out-of-scope request (a cookie recipe) to test refusal/redirection behaviour.

Why identical generation settings for A and B? `config.GENERATION_KWARGS` (greedy decoding, `max_new_tokens=300`, `repetition_penalty=1.1`) is applied to *both* models so any quality difference reflects the models themselves, not different decoding strategies.

Imports and Environment Setup

In [1]:
```
import sys, os, json, time
sys.path.insert(0, os.path.abspath(".."))

# If running in the Virtual Lab for the first time, uncomment:
# %pip install -q transformers accelerate peft bitsandbytes torch datasets

import torch
from src.config import (SLM_MODEL_ID, REFERENCE_LLM_MODEL_ID, GENERATION_KWARGS,
                        SYSTEM_PROMPT, BENCHMARK_PATH, RESULTS_DIR)
from src.data_utils import load_benchmark, format_chat_prompt
from src.model_utils import load_causal_lm, generate_response
from src.eval_utils import save_json

print("CUDA available:", torch.cuda.is_available())
if torch.cuda.is_available():
    print("GPU:", torch.cuda.get_device_name(0))
    print("VRAM (GB):", round(torch.cuda.get_device_properties(0).total_memory /
else:
    print("WARNING: no GPU detected -- this notebook must be run on the Virtual
```

```
CUDA available: True
GPU: NVIDIA GeForce RTX 4050 Laptop GPU
VRAM (GB): 6.4
```

Load the Frozen Benchmark (built in Notebook 01)

In [2]:
```
benchmark = load_benchmark(BENCHMARK_PATH)
assert len(benchmark) == 8, "PS2 requires exactly 8 frozen benchmark prompts"
for item in benchmark:
    print(f"[{item['id']}] ({item['type']}) {item['prompt']}")
```

```
[B1] (in_domain_factual) How do I activate a new debit card I just received from
NovaBank?
[B2] (in_domain_factual) What documents do I need to apply for a home loan with N
ovaBank?
[B3] (in_domain_factual) An ATM swallowed my card. What should I do?
[B4] (formatting_sensitive) Explain how to reset my online banking password. Resp
ond ONLY as a numbered list of exactly 4 steps, no extra text before or after the
list.
[B5] (reasoning) I have $500 in my checking account. I want to transfer $300 to s
avings and then pay a $250 phone bill from checking. Do I have enough balance for
both, and in what order should I do them?
[B6] (out_of_scope) Can you give me a recipe for chocolate chip cookies?
[B7] (in_domain_factual) What is the difference between blocking a credit card an
d cancelling it?
[B8] (in_domain_factual) I want to dispute an ATM withdrawal that I did not make.
Walk me through the process.
```

System (A): Base SLM -- No Fine-Tuning

`Qwen/Qwen2.5-1.5B-Instruct` loaded in bf16 (no quantisation needed at 1.5B -- fits in ~3 GB, and we want (A)'s *unmodified* out-of-the-box quality, matching how it will later

be fine-tuned in Notebook 03 so the before/after comparison in Notebook 05 is not confounded by a precision change).

In [3]:
```
slm_model, slm_tokenizer = load_causal_lm(SLM_MODEL_ID, quantize_4bit=False)
print(f"Loaded {SLM_MODEL_ID}")
print(f"Parameters: {sum(p.numel() for p in slm_model.parameters()):,}")
```

```
C:\Users\hamza\.virtualenvs\Banking-Assistant-SLM-Finetuning-koZdX-BY\Lib\site-packages\huggingface_hub\constants.py:310: FutureWarning: The `HF_HUB_ENABLE_HF_TRANSFER` environment variable is deprecated as 'hf_transfer' is not used anymore. Please use `HF_XET_HIGH_PERFORMANCE` instead to enable high performance transfer with Xet. Visit https://huggingface.co/docs/huggingface_hub/package_reference/environment_variables#hfxethighperformance for more details.
  warnings.warn(
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
C:\Users\hamza\.virtualenvs\Banking-Assistant-SLM-Finetuning-koZdX-BY\Lib\site-packages\accelerate\utils\modeling.py:817: UserWarning: expandable_segments not supported on this platform (Triggered internally at C:\actions-runner\_work\pytorch\pytorch\c10/cuda/CUDAAllocatorConfig.h:40.)
  _ = torch.tensor([0], device=i)
Loaded Qwen/Qwen2.5-1.5B-Instruct
Parameters: 1,543,714,304
```

In [4]:
```
responses_A = {}
for item in benchmark:
    prompt_text = format_chat_prompt(slm_tokenizer, item["prompt"], SYSTEM_PROMP
    t0 = time.time()
    response = generate_response(slm_model, slm_tokenizer, prompt_text, **GENERA
    dt = time.time() - t0
    responses_A[item["id"]] = response
    print(f"=== [{item['id']}] ({item['type']}) -- {dt:.1f}s ===")
    print("Q:", item["prompt"])
    print("A (base SLM):", response)
    print()
```

=== [B1] (in_domain_factual) -- 14.6s ===
Q: How do I activate a new debit card I just received from NovaBank?
A (base SLM): To activate your new debit card at NovaBank, follow these steps:

1. **Review Your Account Information**: Before proceeding, make sure you have all necessary information such as your account number, PIN, and any other required do cuments.

2. **Log in to Your Online Banking or Mobile App**: Go to the NovaBank website or download their mobile app on your device. Log in using your login credentials.

3. **Navigate to the Card Activation Section**: Once logged in, look for an optio n related to card activation within your account dashboard. This might be under "My Accounts," "Card Services," or similar sections.

4. **Select the Debit Card Option**: Choose the option that allows you to activat e your debit card. You may need to select the specific card type if there are mul tiple cards associated with your account.

5. **Enter Required Information**: Follow the prompts to enter your personal iden tification numbers (PIN) and any additional details needed by the system.

6. **Complete Verification Steps**: Depending on the method of verification provi ded by NovaBank, complete the verification process. This could involve answering security questions, entering a code sent to your phone, or providing biometric da ta like fingerprints.

7. **Activate Your Card**: After completing the verification step, proceed to act ivate your debit card. This typically involves confirming that you understand the terms and conditions and agreeing to receive future statements via email or text.

8. **Check Your Statement**: Once activated, check your statement to ensure every thing has been set

=== [B2] (in_domain_factual) -- 8.8s ===
Q: What documents do I need to apply for a home loan with NovaBank?
A (base SLM): To apply for a home loan with NovaBank, you typically need to provi de the following documents:

1. **Identification**: A valid government-issued ID such as a passport or drive r's license.
2. **Proof of Income**: Recent pay stubs or tax returns showing your income over the past two years.
3. **Credit Report**: A copy of your credit report from one of the major credit b ureaus (Equifax, Experian, or TransUnion).
4. **Employment Verification**: Proof of employment, such as a recent letter from your employer confirming your current position and salary.
5. **Home Purchase Agreement**: A signed agreement outlining the terms of the pur chase of your property.
6. **Willingness to Pay**: Evidence that you have the ability to repay the loan, often shown by a down payment on the property.

It’s also advisable to bring these documents in person when applying at a branch, but if you prefer, you can submit them via mail or online through NovaBank's appl ication portal.

=== [B3] (in_domain_factual) -- 8.7s ===
Q: An ATM swallowed my card. What should I do?
A (base SLM): If your ATM swallowed your card, it's important to act quickly:

1. **Contact Your Bank**: Reach out to your bank immediately. They will guide you on what to do next.

2. **Report It to the ATM Manufacturer**: Contact the manufacturer of the ATM where your card was swallowed for further instructions or assistance.

3. **Keep Track of Any Attempts to Retrieve Your Card**: Note any attempts by others to retrieve your card from the machine.

4. **Follow Up**: Keep in touch with both your bank and the ATM manufacturer to ensure that they have taken appropriate action regarding your lost card.

5. **Consider Using an Alternative ATM**: If possible, use another ATM to withdraw money until your card is retrieved.

Remember, it’s crucial to follow these steps as soon as possible to minimize potential risks and losses.

=== [B4] (formatting_sensitive) -- 2.5s ===
Q: Explain how to reset my online banking password. Respond ONLY as a numbered list of exactly 4 steps, no extra text before or after the list.
A (base SLM): 1. Log in to your NovaBank account.
2. Click on "My Account" or "Profile."
3. Look for an option labeled "Change Password" or similar.
4. Follow the instructions provided by clicking "Reset Password."

=== [B5] (reasoning) -- 6.5s ===
Q: I have $500 in my checking account. I want to transfer $300 to savings and then pay a $250 phone bill from checking. Do I have enough balance for both, and in what order should I do them?
A (base SLM): Yes, you have enough balance for both transactions. You would first transfer $300 from your checking account to your savings account, leaving you with $200 in your checking account. Then, you could use that remaining $200 to pay your $250 phone bill from your checking account.

Here's how it works:
1. Transfer $300 from your checking account to your savings account.
2. Use the remaining $200 in your checking account to cover the $250 phone bill.

Remember to check if there are any transaction limits or restrictions on transferring funds between accounts at NovaBank.

=== [B6] (out_of_scope) -- 2.9s ===
Q: Can you give me a recipe for chocolate chip cookies?
A (base SLM): I'm sorry, but I can't assist with that. My capabilities are limited to providing information related to NovaBank's banking services such as cards, loans, mortgages, accounts, transfers, ATMs, passwords, and fees. I don't have the ability to provide recipes or any other non-financial content.

```
=== [B7] (in_domain_factual) -- 7.1s ===
Q: What is the difference between blocking a credit card and cancelling it?
A (base SLM): Blocking a credit card involves temporarily preventing transactions
from being processed on your account while keeping the card itself active for fut
ure use. This action allows you to manage your spending more effectively by stopp
ing unauthorized charges or fraudulent activities.

Cancelling a credit card means permanently ending all access to that particular c
ard. Once cancelled, the card will no longer be usable for any transactions, incl
uding those made in person at an ATM or online through our website or mobile app.
Cancelling a card also removes its associated security features like PIN protecti
on, which may need to be reactivated if you decide to reopen the card later.

Both actions have different implications depending on how they affect your financ
ial situation and personal preferences regarding card usage. Blocking helps maint
ain control over your funds during emergencies, whereas cancellation completely s
tops using the card altogether.

=== [B8] (in_domain_factual) -- 12.6s ===
Q: I want to dispute an ATM withdrawal that I did not make. Walk me through the p
rocess.
A (base SLM): Certainly! To dispute an ATM withdrawal that you didn't make, follo
w these steps:

1. **Review Your Statement**: First, check your recent bank statement for any tra
nsactions that might have been made at an ATM or another location where withdrawa
ls were processed.

2. **Contact Your Bank**: Reach out to your bank's customer service department. Y
ou can usually do this by phone, email, or online chat. Provide them with details
of the transaction, including the date, amount, and location.

3. **File a Dispute**: Once you've identified the incorrect transaction, file a f
ormal dispute with your bank. This will typically involve providing documentation
such as your account information, the disputed transaction details, and possibly
a copy of your identification.

4. **Follow Up**: After submitting your dispute, keep track of any communications
from your bank regarding their investigation. It’s important to stay in touch wit
h them throughout the process.

5. **Wait for Resolution**: The bank has 60 days to investigate the dispute and e
ither agree or disagree with it. During this time, they may contact you to gather
more information or ask for additional documents.

6. **Receive a Response**: Depending on how quickly your bank processes disputes,
you should receive a response within the 60-day period. If your bank agrees that
the transaction was fraudulent, they will reverse it and issue you a refund.

7. **Keep Records**: Throughout the entire process, maintain all correspondence a
nd records related to the dispute. These documents will
```

Free SLM memory before loading the (larger) reference LLM

In [5]:
```python
del slm_model
import gc
```

```
gc.collect()
if torch.cuda.is_available():
    torch.cuda.empty_cache()
print("SLM unloaded.")
```

```
SLM unloaded.
```

System (B): Reference LLM -- No Fine-Tuning

`Qwen/Qwen2.5-7B-Instruct`, loaded with 4-bit NF4 quantisation (bitsandbytes) for **inference only** -- it is never fine-tuned, per PS2's rules for the reference model. Same family as the SLM (isolates the effect of scale) and ~4.7x more parameters, satisfying PS2's "substantially larger, 7B+" requirement.

If your Virtual Lab GPU cannot fit even the 4-bit 7B model, see `README.md` -- "Reference LLM alternatives" for a smaller (but still 7B+ where possible) or API-based fallback; do not silently swap to a <7B model without updating the report's model-choice justification.

In [6]:
```
ref_model, ref_tokenizer = load_causal_lm(REFERENCE_LLM_MODEL_ID, quantize_4bit=
print(f"Loaded {REFERENCE_LLM_MODEL_ID} (4-bit)")
print(f"Parameters: {sum(p.numel() for p in ref_model.parameters()):,}")
```

```
W0811 23:41:16.025000 26200 Lib\site-packages\torch\utils\flop_counter.py:29] tri
ton not found; flop counting will not work for triton kernels
Loaded Qwen/Qwen2.5-7B-Instruct (4-bit)
Parameters: 4,352,972,288
```

In [7]:
```
responses_B = {}
for item in benchmark:
    prompt_text = format_chat_prompt(ref_tokenizer, item["prompt"], SYSTEM_PROMP
    t0 = time.time()
    response = generate_response(ref_model, ref_tokenizer, prompt_text, **GENERA
    dt = time.time() - t0
    responses_B[item["id"]] = response
    print(f"=== [{item['id']}] ({item['type']}) -- {dt:.1f}s ===")
    print("Q:", item["prompt"])
    print("A (reference LLM):", response)
    print()
```

=== [B1] (in_domain_factual) -- 13.7s ===
Q: How do I activate a new debit card I just received from NovaBank?
A (reference LLM): To activate your new NovaBank debit card, follow these steps:

1. **Call Customer Service**: Dial 800-526-3226 to speak with a representative wh
o can guide you through the activation process over the phone.

2. **Visit a Branch**: You can also visit any NovaBank branch where a teller will
assist you in activating your card.

3. **Online Activation**: Log into your online banking account at novabank.com. L
ook for the "Card Services" or "Debit Card Services" section and follow the instr
uctions provided there to activate your card.

4. **Mobile App**: If you have downloaded the NovaBank mobile app, open it and lo
ok for options related to card management or services. There should be an option
to activate your new card within the app.

Once activated, you can start using your card for transactions. Remember to keep
your PIN confidential and change it if you haven't done so already by following t
he prompts during the activation process.

=== [B2] (in_domain_factual) -- 12.4s ===
Q: What documents do I need to apply for a home loan with NovaBank?
A (reference LLM): To apply for a home loan with NovaBank, you will typically nee
d to provide the following documents:

1. **Proof of Income**: W-2 forms, pay stubs, or tax returns from the past two ye
ars.
2. **Identification**: A government-issued ID such as a driver's license or passp
ort.
3. **Employment Verification**: Letter from your current employer or recent emplo
yment history.
4. **Credit Report Authorization**: Consent to allow NovaBank to pull your credit
report.
5. **Bank Statements**: Recent bank statements showing your financial activity.
6. **Down Payment Savings**: Documentation of funds available for the down paymen
t and closing costs.
7. **Property Information**: Proof of property ownership if applicable, or detail
s about the property you plan to purchase.

It’s best to check directly with NovaBank for their specific requirements, as the
y may have additional or different documentation needs depending on your individu
al circumstances.

=== [B3] (in_domain_factual) -- 10.3s ===
Q: An ATM swallowed my card. What should I do?
A (reference LLM): If an ATM has swallowed your card, here’s what you should do:

1. **Stay Calm**: Do not panic.
2. **Check for the Card**: Look around the ATM area to see if your card is nearby or on the ground.
3. **Contact NovaBank**: Call our customer service line at [insert phone number] to report the issue. They will guide you through the next steps.
4. **Follow Up**: Keep track of any communication from the bank regarding your card retrieval.
5. **Secure Your Account**: Consider changing your PIN or updating your account details as a precaution.

We apologize for any inconvenience this may have caused. Please let us know if you need further assistance!

=== [B4] (formatting_sensitive) -- 3.5s ===
Q: Explain how to reset my online banking password. Respond ONLY as a numbered list of exactly 4 steps, no extra text before or after the list.
A (reference LLM): 1. Log in to your NovaBank Online Banking account.
2. Click on "Settings" or "Profile" in the menu.
3. Select "Change Password" or similar option.
4. Follow the prompts to set a new password.

=== [B5] (reasoning) -- 21.1s ===
Q: I have $500 in my checking account. I want to transfer $300 to savings and the
n pay a $250 phone bill from checking. Do I have enough balance for both, and in
what order should I do them?
A (reference LLM): To ensure you have enough balance for both transactions, let's
break it down:

1. **Transfer $300 from Checking to Savings:**
   - You currently have $500 in your checking account.
   - After transferring $300 to savings, your checking account will have:
     \[
     \$500 - \$300 = \$200
     \]

2. **Pay $250 Phone Bill from Checking:**
   - After the transfer, you have $200 left in your checking account.
   - Paying a $250 phone bill would require:
     \[
     \$200 - \$250 = -\$50
     \]
   - This results in an overdraft of $50.

Conclusion:
- Yes, you have enough balance to perform the first transaction (transfer $300 to
savings), but not the second one (pay $250 phone bill) because after the transfe
r, you'll be short by $50.

Order of Transactions:
- It doesn't matter which transaction you do first as long as you don't exceed yo
ur available balance before performing any transaction. However, if you need to p
rioritize, you might consider paying the phone bill last or finding another way t
o cover the shortfall.

If you proceed with the transfer first, you must ensure that you have at least $2
50 remaining in your checking account when you

=== [B6] (out_of_scope) -- 3.2s ===
Q: Can you give me a recipe for chocolate chip cookies?
A (reference LLM): I'm sorry, but I can only provide information related to NovaB
ank's banking services. For a recipe for chocolate chip cookies, you might want t
o check out cooking websites or food blogs that offer recipes. Enjoy baking!

=== [B7] (in_domain_factual) -- 14.6s ===
Q: What is the difference between blocking a credit card and cancelling it?
A (reference LLM): At NovaBank, blocking a credit card and canceling a credit car
d serve different purposes:

1. **Blocking a Credit Card**: When you block a credit card, it temporarily suspe
nds its use until you unblock it. This means that no transactions can be processe
d through the blocked card during this period. You might choose to block your car
d if you suspect fraudulent activity or if you need to take a break from using th
e card for personal reasons.

2. **Canceling a Credit Card**: Canceling a credit card permanently ends its func
tionality and removes it from your account. Once canceled, the card cannot be use
d for any transactions, and the account associated with the card will close. Canc
ellation typically involves returning the physical card and updating your records
to reflect the change in status.

If you need to either block or cancel your credit card, please contact our custom
er service team at [phone number] or log into your online banking account to mana
ge your card settings.

=== [B8] (in_domain_factual) -- 19.3s ===
Q: I want to dispute an ATM withdrawal that I did not make. Walk me through the p
rocess.
A (reference LLM): Certainly! Here’s how you can dispute an unauthorized ATM with
drawal from NovaBank:

1. **Contact Us Immediately**: Call our Customer Service line at [insert phone nu
mber] as soon as possible to report the unauthorized transaction.

2. **Provide Details**: When you call, be prepared to provide details such as:
   - The date of the transaction.
   - The amount withdrawn.
   - Any account numbers or card numbers related to the transaction.
   - A brief description of why you believe it was unauthorized.

3. **Follow Up in Writing**: After your initial call, follow up by sending a writ
ten statement to us via certified mail or email. Include all relevant information
and any supporting documents like receipts or statements.

4. **Review Your Account**: Keep an eye on your account for any further transacti
ons or communications regarding this issue. You may receive a fraud alert or addi
tional steps to take.

5. **Secure Your Account**: Consider changing your password or PIN if you haven’t
already done so. This helps prevent future unauthorized access.

6. **Stay Informed**: Stay informed about any updates from NovaBank regarding the
status of your dispute. They will inform you once they have investigated the matt
er.

If you need more assistance or have other questions about banking services, feel
free to ask!

Save Verbatim Responses (required deliverable for Task 2)

In [8]:

```
baseline_results = {
    "benchmark": benchmark,
    "system_A_base_slm": {"model_id": SLM_MODEL_ID, "responses": responses_A},
    "system_B_reference_llm": {"model_id": REFERENCE_LLM_MODEL_ID, "responses":
    "generation_kwargs": GENERATION_KWARGS,
}
save_json(baseline_results, os.path.join(RESULTS_DIR, "task2_baseline_responses.
print("Saved results/task2_baseline_responses.json")

import pandas as pd
rows = []
for item in benchmark:
    rows.append({
        "id": item["id"], "type": item["type"], "prompt": item["prompt"],
        "A_base_slm": responses_A[item["id"]],
        "B_reference_llm": responses_B[item["id"]],
    })
side_by_side = pd.DataFrame(rows)
side_by_side.to_csv(os.path.join(RESULTS_DIR, "task2_side_by_side.csv"), index=F
side_by_side
```

```
Saved results/task2_baseline_responses.json
```

Out[8]:

	id	type	prompt	A_base_slm	B_reference_llm
0	B1	in_domain_factual	How do I activate a new debit card I just rece...	To activate your new debit card at NovaBank, f...	To activate your new NovaBank debit card, foll...
1	B2	in_domain_factual	What documents do I need to apply for a home l...	To apply for a home loan with NovaBank, you ty...	To apply for a home loan with NovaBank, you wi...
2	B3	in_domain_factual	An ATM swallowed my card. What should I do?	If your ATM swallowed your card, it's importan...	If an ATM has swallowed your card, here's what...
3	B4	formatting_sensitive	Explain how to reset my online banking passwor...	1. Log in to your NovaBank account.\n2. Click ...	1. Log in to your NovaBank Online Banking acco...
4	B5	reasoning	I have $500 in my checking account. I want to ...	Yes, you have enough balance for both transact...	To ensure you have enough balance for both tra...
5	B6	out_of_scope	Can you give me a recipe for chocolate chip co...	I'm sorry, but I can't assist with that. My ca...	I'm sorry, but I can only provide information ...
6	B7	in_domain_factual	What is the difference between blocking a cred...	Blocking a credit card involves temporarily pr...	At NovaBank, blocking a credit card and cancel...
7	B8	in_domain_factual	I want to dispute an ATM withdrawal that I did...	Certainly! To dispute an ATM withdrawal that y...	Certainly! Here's how you can dispute an unaut...

Limitations / Gap Table

Filled in from the actual outputs above (`results/task2_baseline_responses.json` , run locally on an RTX 4050 Laptop GPU -- see report Appendix C for why this ran outside the Virtual Lab).

Dimension	Where SLM (A) failed vs. LLM (B)	Evidence (prompt id)
Domain knowledge	A gives generic, non-NovaBank-branded procedures with no concrete channel details (no phone number, no branch/app names); B consistently grounds answers in NovaBank-specific channels (a phone number, novabank.com, "our" branding)	B1, B3
Instruction / format adherence	No gap at baseline: A already produced exactly 4 numbered steps with no extra text for the strict-format prompt	B4 (both correct)

Dimension	Where SLM (A) failed vs. LLM (B)	Evidence (prompt id)
Completeness / reasoning	A failed the arithmetic reasoning task outright: it states "you have enough balance for both" and then describes paying a $250 bill from a 200 remaining balance -- an internal contradiction -- and never flags the resulting $50 overdraft. B performs the subtraction explicitly (500-300 =200, then 200−250=-$50) and correctly concludes there isn't enough balance for both	B5
Hallucination	A states a specific "60 days" dispute-resolution window with confident, unhedged phrasing but no NovaBank source backs that number; B avoids committing to an unverified specific timeframe	B8
Persona / brand voice consistency	A rarely speaks as NovaBank itself ("your bank", "the ATM manufacturer") despite the system prompt instructing it to be the NovaBank assistant; B consistently uses "our", "NovaBank", and first-person service-desk phrasing	B1, B3, B7
Out-of-scope handling	No gap at baseline: both A and B correctly declined the off-topic cookie-recipe request	B6 (both correct)

Inference: The baseline gap is smaller than expected on refusal and strict-format compliance (B4, B6) -- the 1.5B model already handles those. It is concentrated instead where it matters most: A's arithmetic/logical reasoning (B5) isn't just weaker, it's outright wrong and self-contradictory, and A consistently fails to sound like a NovaBank agent rather than a generic assistant. This makes reasoning correctness and consistent NovaBank persona/branding the clearest fine-tuning targets -- format-following and scope-refusal are close to solved already and don't need much training signal.

Conclusion

- Ran the frozen 8-prompt benchmark against (A) the unmodified `Qwen/Qwen2.5-1.5B-Instruct` and (B) the unmodified `Qwen/Qwen2.5-7B-Instruct` (4-bit, inference-only), using identical generation settings.
- Saved verbatim responses to `results/task2_baseline_responses.json` and `results/task2_side_by_side.csv` -- required inputs for Notebook 04 (diagnosis) and Notebook 05 (three-way comparison).
- Completed the qualitative limitations/gap table above from the real outputs.

Next notebook: `03_finetuning_v1.ipynb` -- LoRA fine-tunes the SLM on the Notebook-01 training split to produce Version 1 (v1), with the mandatory efficiency evidence (trainable-parameter % and peak training memory).